SSU Data Dash 2026: Toronto Home-Price Prediction Challenge

The problem

When a home in Toronto goes on the market, everyone wants to know one thing: what will it actually sell for? Buyers, sellers, banks and tax assessors all lean on automated valuation models (AVMs) to answer that. Your team will build one, and test on sales it has never seen.

There are two different questions a pricing model can answer.

- **Off-market: what is this home worth?** You value a home before it is listed, so there is no asking price. This is the hard question, and it counts double.
- **List-aware: what will this listing close at?** The home is on the market and you know its list price. The list price already carries most of what the market knows, so this is much easier.

Pitfalls:

1. Mixing the questions up. A model that uses the list price looks far more accurate than one that does not. Reporting a list-aware result as an answer to the off-market question overstates what the model can do.
2. Time leakage. A model that learns from sales after the one it predicts looks excellent in testing and fails in use. A random train/test split does exactly this. Always train on the past and test on the future.

Your task

Start from the working pipeline we give you and make it better: more accurate, more honest about its uncertainty, or both. Strong submissions might:

- build better features: location, comparable recent sales, market trends, the listing descriptions;
- try other models or combine them;
- estimate uncertainty, not just a single price;
- find where the model fails (by price, property type, neighbourhood) and explain or fix it.

You will predict prices for homes that sold June to September 2026, and we score those predictions against the real (held-back) sale prices after the deadline.

The data

All data is synthetic: no real listing, address or price appears. It was generated from a model calibrated to the Toronto market (property-type price levels, neighbourhoods with their own levels and trends, the 2021 to 2026 boom and correction, sold-to-list ratios), so modelling moves that help on real data help here too.

Download it from huggingface.co/datasets/Baliyang/SSU_data_dash (the starter code downloads it for you).

File	Rows	What it is
`listings`	150,000	Sales from January 2021 to May 2026, with sold prices. Train on these.
`test_off_market`	3,974	Homes to value before listing: no list price. Predict their sale price.
`test_list_aware`	4,064	Listed homes with their list price. Predict their sale price.

Each file comes as `.parquet` and `.csv`. The two test files are different homes.

Group	Columns
Target	`sold_price` (listings only)
Listing	`list_price` (not in `test_off_market`), `date_listed`, `date_sold` and `days_on_market` (listings only)
Property	`house_type_name` (6 types), `house_style`, `bedrooms`, `washrooms`, `parking_total`, `garage`, `house_area` (sq ft, sometimes a range), `lot_front`, `lot_depth` (ft)

Group	Columns
Location	`municipality`, `community` (neighbourhood), `latitude`, `longitude`
Text	`remarks`: a listing description, written by a language model
Third-party estimate	`external_avm_estimate`, `external_avm_estimate_date` (listings only)
ID	`ml_num`

Two things to know:

- Location is synthetic. Neighbourhood names, coordinates and streets do not match real maps, so outside map or census data will not line up.
- The third-party estimate. It appears only in `listings`, not in the test files. A real estimate is only usable if it existed before the sale: check `external_avm_estimate_date` against the sale date before you rely on it, and think about whether your model could have it at prediction time.

Your starting point

The starter repository, github.com/erwinzhang7/SSU-Data-Dash, runs end to end. A Colab notebook and an R version are linked from its README.

```
pip install -r requirements.txt
python run.py validate   # walk-forward test on December 2025 to May 2026
python run.py report     # results table and a figure for your report
python run.py predict    # write both submission files
python run.py check      # check them before you hand in
```

Edit `features.py` (features) and `models.py` (models). `validate` predicts each month from the sales before it, the same way we test you. Its numbers are the ones to report.

The bar. On the starter's walk-forward validation (December 2025 to May 2026):

Track	Model	MdAPE	MAPE
Both	Maximum error: a simple baseline, the median sale of the same type in the same neighbourhood over the previous 12 months	17.7%	22.0%
Off-market	Starter baseline: median of the 10 nearest earlier sales	20.8%	27.4%
List-aware	Starter baseline: the list price	7.3%	8.1%

MdAPE is the median absolute percentage error, |predicted − sold| / sold; MAPE is the mean. To earn accuracy points on a track, your hidden-test MdAPE must be below the maximum error: the simple baseline's error on the same hidden homes. The starter's off-market baseline is above it, so you have to improve on it. A gradient-boosted model (`FeatureModel`) is ready to switch on in `models.py`.

What to submit

By **November 8, 11:59 pm EST**, through the form we will email and post on Discord: one PDF report (4 pages max) that links to:

1. A video, 5 minutes max, on YouTube (unlisted is fine, private is not). Judges stop at 5:00.
2. A GitHub repository (public, or shared with the organisers) holding your code and two files, written by `python run.py predict`:

```
submissions/submission_off_market.csv    one row per home in test_off_market
submissions/submission_list_aware.csv    one row per home in test_list_aware

ml_num,predicted_price
SYN-0123456,815000
```

Use the report template in the starter (`report/report.tex`; it compiles on Overleaf). Put every team member's full name on it. Report your walk-forward validation MdAPE for each track: we compare it with your hidden-test error. Late submissions are not reviewed.

How you are judged

The full rubric is in the starter repository. In short:

Round 1, online review (November 9 to 15).

Criterion	Points	What earns it
Accuracy on the hidden test	30	Off-market 20, list-aware 10. Ranked by MdAPE; MAPE breaks ties.
Method and validation	25	Train on the past, test on the future. Nothing from after the sale or from the test set.
Insight and error analysis	20	Where the model fails (price, type, neighbourhood) and why. Honest uncertainty.
Honesty about performance	10	Your validation error should predict your hidden-test error.
Communication	10	A clear five-minute story.
Reproducibility	5	We can rerun your code and get your predictions.

A smaller error that you cannot explain loses to a slightly larger one with a clean, honest method.

Round 2, finale (in person, November 20). The top five teams present. Scores start fresh:

Criterion	Points
The problem and who it serves: a real user, a real decision	25
Recommendations that follow from your results	25
Risks and limits: when not to trust the model	20
Technical credibility under questions	15
Presentation and Q&A	15

Timeline

Date	What
October 8	Case and data released. Workshop, 1 to 3 pm, room 9199, 700 University Ave.
November 8, 11:59 pm EST	Submissions due.
November 9 to 15	Round 1 review.
November 16	Finalists announced.
November 20	Finale, in person, followed by a networking event. Everyone who submits is invited.

Prizes, per team: $900 first, $600 second, $300 third.

Rules

- Outside data only if it covers May 2026 or earlier. List every source with its URL in your report.
- No real listing or sales records. No attempts to recover the hidden test prices. Either disqualifies the team.
- Every number in your video comes from your own validation.
- The off-market model may not use the list price.

Getting help

Join the Discord: discord.gg/TE2UPPj6G. Post in the forum that fits:

- case-questions: the problem, the data, the rules, judging;
- code-help: the starter code, setup, errors;
- workshop-questions: anything from the workshop.

Search before you post: someone may already have asked. Answers posted there apply to every team.

Common mistakes

- A random split. It lets the model see the future. Use `run.py validate` or the same walk-forward idea.
- List price in the off-market model. The off-market test file has none.
- Features built from the whole dataset, such as a neighbourhood average that includes later sales. Compute them from earlier sales only.
- The third-party estimate without checking its date.
- Reporting the best of many tries on the validation months as if it were a fresh test. Your hidden-test error will be worse, and the honesty score will show it.